7 **(a)** Fig. 7.1 shows the electric field pattern around point X.

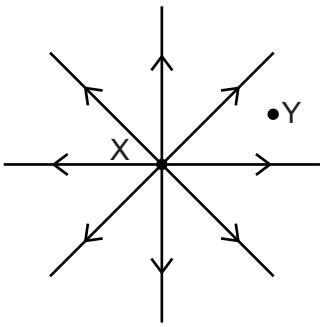


Fig. 7.1

(i) On Fig. 7.1, draw an arrow to indicate the direction of the force on a negative point charge placed at point Y. [2]

(ii) State what is at point X to produce the field pattern shown in Fig. 7.1.

.....

..... [2]

(b) A piece of plastic is charged positively by friction.

State what charge transfers occur during this process.

.....

.....

..... [2]

(c) Explain how the structure of an electrical conductor differs from the structure of an electrical insulator.

.....

..... [2]

[Total: 8]

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